Aaron Daniel Kartmann

Experimental Condensed Matter Physicist

MSc Physics, specialization in Quantum Matter | Radboud University Nijmegen

LinkedIn GitHub Website Nijmegen, The Netherlands

RESEARCH PROFILE

Experimental physicist with an MSc in Physics, specializing in Quantum Matter, from Radboud University Nijmegen. My research experience includes electronic transport and torque magnetometry at high magnetic fields and cryogenic temperatures, as well as laser-based magneto-optical measurements on magnetic thin films. I am interested in experimental and computational research on optics, quantum materials, and nanoscale devices, particularly where a fundamental understanding of physical properties can contribute to future technologies.

RESEARCH EXPERIENCE

Master's Research Project **Sep 2025 – Aug 2026**

Correlated Electron Systems group, HFML-FELIX, Nijmegen

Grade: **9/10**

- Investigated the magnetoresistance, superconductivity, and quantum oscillations of quantum materials using electronic transport measurements and cantilever torque magnetometry at high magnetic fields and cryogenic temperatures.
- Compared the magnetoresistance and Fermi surface of pristine $PtBi_2$ with Pb-doped $PtBi_2$. Found evidence for 2D superconductivity in pristine $PtBi_2$.
- Showed that both compressive and tensile uniaxial strain can tune the magnetoresistance and Fermi surface of WTe_2.
- Investigated the magnetoresistance of $MnSb_2Te_4$. Found an unusual low-temperature upturn in the electrical resistance.
- Gained laboratory experience with superconducting and Bitter magnets, cryogenic systems, vacuum equipment, and sample preparation.
- Developed skills in experimental setup, measurement troubleshooting, data analysis, and interpretation of experimental results.

Supervisor: Dr. Steffen Wiedmann.

Bachelor's Research Project **Feb 2024 – Jul 2024**

Spectroscopy of Solids and Interfaces group, Radboud University Nijmegen

Grade: **8.5/10**

- Demonstrated helicity-dependent all-optical switching (HD-AOS) of magnetization in Pt/Co/Pt thin films on a 50 nm-thick silicon nitride membrane, extending the established effect from conventional glass substrates to a novel thin membrane platform.
- Compared the laser fluence thresholds for HD-AOS of samples on the silicon nitride and glass substrates.
- Gained laboratory experience in optics and working with lasers, specifically Faraday microscopy and Faraday rotation measurements.

Supervisor: Prof. Theo Rasing.

EDUCATION

MSc Physics, specialization in Quantum Matter **2024 – 2026**

Radboud University Nijmegen

Final grade: **8.8/10, Cum Laude**.

BSc Physics and Astronomy **2021 – 2024**

Radboud University Nijmegen

Final grade: **8.3/10, Cum Laude**.

TECHNICAL SKILLS

Programming	Python (data analysis, numerical simulations, machine learning); C++ (object-oriented programming, computationally efficient programming, Python integration)
Experimental Techniques	Laser-based magneto-optical measurements, electronic transport, cantilever torque magnetometry
Laboratory Skills	Cryogenic systems (^{4}He, ^{3}He, liquid nitrogen), vacuum equipment, sample preparation, electrical contacting, thin-film deposition by vacuum evaporation
Software and Tools	OriginPro, LaTeX, Linux, Git, Microsoft Office, HTML, CSS

EXTRACURRICULAR ACTIVITIES

Teaching Assistant, Introduction to Thermodynamics (Warmteleer) **2023/24 and 2024/25**
Radboud University Nijmegen

- Led tutorials, guided first-year students through physics exercises, answered questions, and graded assignments.
- Developed experience in explaining physical concepts clearly.

Website Commission **Sep 2022 – Jul 2023**
Study Association Marie Curie Nijmegen

- Contributed to maintaining the study association's website by fixing issues

LANGUAGES

German: C2
English: C1, Cambridge C1 Advanced **(Jun 2021)**
Dutch: C1, RU-NT2-exam (B2 level) **(Aug 2021)**
French: A1